

Python Lists, Tuples & Dictionaries Quiz

1. How do you create an empty set?

- A) {}
- B) set()
- C) []
- D) ()

Correct Answer: B

Explanation: {} creates an empty dictionary, not a set.

2. What is the output of {1, 2, 2, 3}?

- A) {1, 2, 2, 3}
- B) {1, 2, 3}
- C) {1, 3}
- D) Error

Correct Answer: B

Explanation: Sets automatically remove duplicates.

3. Which operator finds the union of two sets?

- A) &
- B) |
- C) -
- D) ^

Correct Answer: B

Explanation: | combines elements from both sets.

4. What does {1, 2} - {2, 3} return?

- A) {1}
- B) {3}
- C) {1, 3}
- D) {2}

Correct Answer: A

Explanation: - returns elements only in the first set.

5. Which checks if 1 is in set s?

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- A) `s.contains(1)`
- B) `1 in s`
- C) `s.in(1)`
- D) `s.has(1)`

Correct Answer: B

Explanation: `in` is the membership operator.

6. Which method adds multiple elements to a set?

- A) `.add()`
- B) `.update()`
- C) `.insert()`
- D) `.extend()`

Correct Answer: B

Explanation: `.update()` takes an iterable.

7. What is the difference between `.remove()` and `.discard()`?

- A) `.discard()` raises an error if the element is missing
- B) `.remove()` raises an error if the element is missing
- C) No difference
- D) `.discard()` works only for frozen sets

Correct Answer: B

Explanation: Both remove elements, but `.remove()` raises `KeyError` if absent.

8. What does `{1, 2}.isdisjoint({3, 4})` return?

- A) `True`
- B) `False`
- C) `None`
- D) `{1, 2, 3, 4}`

Correct Answer: A

Explanation: The sets share no common elements.

9. Which method returns a copy of a set?

- A) `.clone()`

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- B) `.copy()`
- C) `set.copy()`
- D) `=`

Correct Answer: B

Explanation: `.copy()` creates a shallow copy.

10. What is the output of `{1, 2, 3}.pop()`?

- A) 1
- B) 3
- C) Arbitrary element
- D) Error

Correct Answer: C

Explanation: `.pop()` removes and returns an arbitrary element (order not guaranteed).

11. What does `{1, 2} & {2, 3}` return?

- A) `{1, 2, 3}`
- B) `{2}`
- C) `{1, 3}`
- D) `set()`

Correct Answer: B

Explanation: `&` returns the intersection (common elements).

12. Which operation gives elements in either set but not both?

- A) Union (`|`)
- B) Intersection (`&`)
- C) Symmetric difference (`^`)
- D) Difference (`-`)

Correct Answer: C

Explanation: `^` returns elements in exactly one set.

13. What does `{1, 2} <= {1, 2, 3}` check?

- A) Proper superset
- B) Subset

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C) Disjoint sets

D) Equality

Correct Answer: B

Explanation: `<=` checks if the first set is a subset of the second.

14. `{1, 2} | {3}` is equivalent to:

A) `{1, 2}.union([3])`

B) `{1, 2}.update({3})`

C) Both A and B

D) Neither

Correct Answer: A

Explanation: `|` and `.union()` both perform unions, but `.update()` modifies the original set.

15. What is the output of `len({1, 2, 2, 3})`?

A) 4

B) 3

C) 2

D) Error

Correct Answer: B

Explanation: Duplicates are ignored in sets.

16. What is a frozenset?

A) A sorted set

B) An immutable set

C) A set with duplicate elements

D) A set that only stores numbers

Correct Answer: B

Explanation: Frozensets cannot be modified after creation.

17. Which is a valid set element?

A) `[1, 2]`

B) `(1, 2)`

C) `{1: 2}`

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D) {1, 2}

Correct Answer: B

Explanation: Only immutable types (like tuples) can be set elements.

18. What does frozenset([1, 2]) return?

A) {1, 2}

B) (1, 2)

C) An immutable {1, 2}

D) Error

Correct Answer: C

Explanation: Converts the list to an unchangeable set.

19. Which is faster for membership testing?

A) Lists

B) Sets

C) Tuples

D) All are equal

Correct Answer: B

Explanation: Sets use hash tables for O(1) lookups.

20. What is the output of {1, 2} == {2, 1}?

A) True

B) False

C) None

D) Error

Correct Answer: A

Explanation: Sets are unordered collections.